

1 Question on Error Calculations

Question: For $f(x) = 7e^{0.5x}$ and $h = 0.3$. Find the approximate value of $f'(x)$ at $x = 2$, the true value of $f'(x)$, and the true error.

Solution:

The approximate value of the derivative is calculated using the forward-difference formula:

$$f'(x) = \frac{f(x+h) - f(x)}{h}.$$

Now, $x = 2$ and $h = 0.3$.

$$f'(2) = \frac{f(2+0.3) - f(2)}{0.3} = \frac{7e^{0.5 \times 2.3} - 7e^{0.5 \times 2}}{0.3}.$$

$$f'(2) = 10.256.$$

Exact value of $f'(x)$:

Given

$$f(x) = 7e^{0.5x},$$

$$f'(x) = 7 \cdot 0.5 e^{0.5x}.$$

So,

$$f'(2) = 7 \times 0.5 e^{0.5 \times 2} = 9.514.$$

True error:

$$E_t = \text{true value} - \text{approximate value} = 9.514 - 10.256 = -0.75061.$$

Question: For $f(x) = 7e^{0.5x}$ and $h = 0.3$. Find the following:

- (a) $f'(2)$ using $h = 0.3$
- (b) $f'(2)$ using $h = 0.15$
- (c) Approximate error for the value of $f'(2)$.

Ans: 0.38

2 Algebraic Equations

An **algebraic equation** is defined by setting an algebraic function equal to zero. These include polynomial equations as a special case, but also equations involving **roots** and **rational terms**.

1. General Polynomial Form (The most common algebraic equation)

The general form of a polynomial equation of degree n :

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

2. Non-Polynomial Algebraic Examples

Rational Algebraic Equation

This involves variables in the denominator.

$$\frac{x^2 - 4}{x + 1} = 5$$

This can be rewritten into a polynomial form after clearing the denominator, $x^2 - 4 = 5(x + 1)$, but is initially presented as a rational equation.

Root (or Radical) Equation

This involves variables under a root sign (fractional exponents).

$$\sqrt{x + 2} = x$$

This can also be converted to a polynomial form by squaring both sides, but it is classified as a radical equation.

Equation with Fractional Exponents

An equation where the variable has non-integer, rational exponents, which is characteristic of an algebraic function.

$$x^{3/2} + 5x^{1/2} - 6 = 0$$

3 Polynomial Equations

A **polynomial equation** of degree n in the variable x sets a polynomial equal to zero.

The general form of a polynomial is:

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

The general polynomial equation is:

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

where $a_n, a_{n-1}, \dots, a_0$ are coefficients and $a_n \neq 0$. These equations have n roots and can have at least

1. real and different
2. real and repeated
3. complex numbers as roots.

Common Examples

Linear Equation (Degree 1)

$$a_1 x + a_0 = 0$$

Quadratic Equation (Degree 2)

$$a_2 x^2 + a_1 x + a_0 = 0$$

Cubic Equation (Degree 3)

$$a_3x^3 + a_2x^2 + a_1x + a_0 = 0$$

The Quadratic Formula (Solution for Degree 2)

The roots of $ax^2 + bx + c = 0$ are given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

4 Transcendental Equations

A **transcendental equation** is an equation that cannot be expressed as a finite-degree polynomial equation. In simpler terms, these are equations involving **transcendental functions** of the variable (like exponential, logarithmic, or trigonometric functions).

Key Distinction: Transcendental vs. Algebraic

- **Algebraic Equation:** Can be written as $P(x) = 0$, where $P(x)$ is a polynomial.
- **Transcendental Equation:** Cannot be reduced to $P(x) = 0$.

Common Examples of Transcendental Equations

Transcendental equations typically require numerical methods (like Newton's method or bisection) for finding approximate solutions, as they generally lack closed-form analytical solutions.

1. Exponential Equations

Equations involving the base of the natural logarithm (e).

$$e^x = x^2$$

$$5^{2x-1} = 10$$

2. Logarithmic Equations

Equations involving logarithms ($\ln$ or $\log$).

$$\ln(x) + x = 5$$

$$\log_{10}(x+2) = 1 - \log_{10}(x)$$

3. Trigonometric Equations

Equations involving sine, cosine, tangent, etc.

$$\sin(x) = x$$

$$\cos(2x) + 3\sin(x) = 0$$

4. Hyperbolic Functions

Equations involving hyperbolic sine or cosine ($\sinh$, $\cosh$).

$$\cosh(x) = \sqrt{x}$$

A transcendental equation may have a finite or an infinite number of real roots or may not have real root at all. **Note:** Closed-form solution means that the solution can be computed using finite steps and exact formula. And a numerical method is not a closed-form solution.

5 The Mean Value Theorem (MVT)

Statement

If a function $f(x)$ satisfies the following conditions on the closed interval $[a, b]$, then there exists at least one number c in the open interval (a, b) such that the derivative at c , $f'(c)$, equals the slope of the secant line connecting the endpoints.

The Conditions

The function $f(x)$ must be:

1. **Continuous** on the closed interval $[a, b]$.
2. **Differentiable** on the open interval (a, b) .

Geometric Interpretation

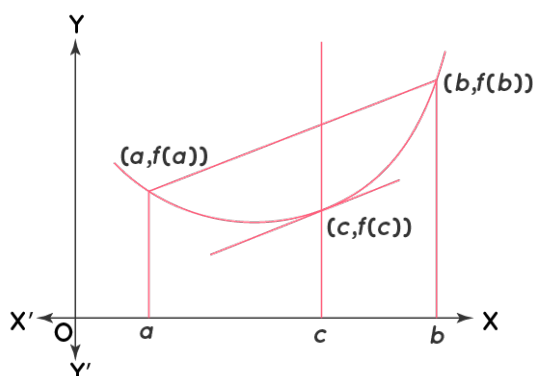


Figure 1: Mean Value Theorem

Suppose f is a continuous function in the closed interval $[a, b]$ and differentiable on (a, b) . We draw a secant line joining the point $(a, f(a))$ to point $(b, f(b))$. We can clearly see a point c where $a < c < b$ such that the tangent line to the graph of f at $x = c$ is parallel to the secant line and has same slope. The slope of the secant line is $\frac{f(b)-f(a)}{b-a}$ and that of the tangent line is $f'(c)$. So,

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Example of the Mean Value Theorem (MVT)

Problem Statement

Verify the Mean Value Theorem for the function $f(x) = x^2 + 2x - 1$ on the interval $[a, b] = [0, 2]$, and find all values of c that satisfy the conclusion of the theorem.

1. Check Conditions

The function $f(x) = x^2 + 2x - 1$ is a polynomial, which is continuous and differentiable everywhere. Thus, the conditions for MVT are satisfied on $[0, 2]$.

2. Calculate Endpoints and Average Slope

The function values at the endpoints are:

$$\begin{aligned}f(a) &= f(0) = (0)^2 + 2(0) - 1 = -1 \\f(b) &= f(2) = (2)^2 + 2(2) - 1 = 4 + 4 - 1 = 7\end{aligned}$$

The average slope (secant line slope) is:

$$\frac{f(b) - f(a)}{b - a} = \frac{f(2) - f(0)}{2 - 0} = \frac{7 - (-1)}{2} = \frac{8}{2} = 4$$

3. Calculate Derivative and Apply MVT Formula

The derivative of the function is:

$$f'(x) = 2x + 2$$

According to the MVT, we set the instantaneous slope equal to the average slope: $f'(c) = 4$.

$$2c + 2 = 4$$

4. Solve for c

Solving the equation for c :

$$\begin{aligned}2c + 2 &= 4 \\2c &= 2 \\c &= 1\end{aligned}$$

5. Conclusion

Since $c = 1$ is in the open interval $(0, 2)$, it satisfies the conclusion of the Mean Value Theorem.

6 Taylor Series

The Taylor Series is an infinite sum that represents a function as an infinite polynomial. It is built from the function's derivatives at a single point (the center of the series).

General Formula

If a function $f(x)$ has derivatives of all orders at a point $x = a$, the Taylor Series for $f(x)$ centered at a is given by:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

Expanded Form

The expanded form of the Taylor Series is:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \dots$$

Key Components:

- a : The **center** (or point of expansion) of the series.
- $f^{(n)}(a)$: The n -th derivative of $f(x)$ evaluated at $x = a$.
- $n!$: n factorial.
- $(x - a)^n$: The power term, centered at a .

Maclaurin Series (Special Case)

The Maclaurin Series is simply a Taylor Series centered at $\mathbf{a} = \mathbf{0}$.

Maclaurin Formula

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

Maclaurin Expanded Form

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

Taylor Polynomial (Approximation)

The N -th degree Taylor Polynomial, $P_N(x)$, is used to approximate the function $f(x)$ by truncating the infinite series after the term of degree N :

$$P_N(x) = \sum_{n=0}^N \frac{f^{(n)}(a)}{n!} (x - a)^n$$

Example: Maclaurin Series Approximation

Problem Statement

Find the value of $e^{0.25}$ using the first five terms of the Maclaurin series.

Solution

The first five terms of the Maclaurin series for e^x can be calculated as follows. We need the function and its first four derivatives evaluated at $a = 0$.

$$\begin{array}{ll} f(x) = e^x & \implies f(0) = e^0 = 1 \\ f'(x) = e^x & \implies f'(0) = e^0 = 1 \\ f''(x) = e^x & \implies f''(0) = e^0 = 1 \\ f'''(x) = e^x & \implies f'''(0) = e^0 = 1 \\ f^{(4)}(x) = e^x & \implies f^{(4)}(0) = e^0 = 1 \end{array}$$

Thus, from the Maclaurin series formula for $f(x)$:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \dots$$

Substituting the values for $f^{(n)}(0)$:

$$e^x \approx 1 + 1x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4$$

$$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$$

Now, substitute $x = 0.25$ into the approximation:

$$e^{0.25} \approx 1 + 0.25 + \frac{(0.25)^2}{2!} + \frac{(0.25)^3}{3!} + \frac{(0.25)^4}{4!}$$

Calculating the terms:

$$\begin{aligned} 1 &= 1.000000 \\ 0.25 &= 0.250000 \\ \frac{(0.25)^2}{2!} &= \frac{0.0625}{2} = 0.031250 \\ \frac{(0.25)^3}{3!} &= \frac{0.015625}{6} \approx 0.002604 \\ \frac{(0.25)^4}{4!} &= \frac{0.00390625}{24} \approx 0.000163 \end{aligned}$$

Summing these terms:

$$e^{0.25} \approx 1.000000 + 0.250000 + 0.031250 + 0.002604 + 0.000163$$

$$e^{0.25} \approx 1.284017$$

Rounding to four decimal places as typically implied in such problems:

$$e^{0.25} \approx 1.2840$$

The exact value of $e^{0.25}$ up to 5 significant digits is 1.2840.

7 Methods of Solving non-linear equations:

For finding the roots of nonlinear equations, 4 main ways are:

1. Direct Analytical Methods
2. Graphical Methods
3. Trial and error Methods
4. Iterative Methods

Direct Analytical Method

For some non-linear equations, there exists formulas for finding the roots of the equations which are known to be direct analytical methods. For example, for a Quadratic equation:

$$ax^2 + bx + c = 0$$

The closed form solution is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

But not all equations can be solved through this way. For example, the equation $2 \sin(x) - x = 0$ cannot be solved analytically. Direct methods are limiting as these methods can be utilized to solve few types of equations in certain simple cases.

Graphical Methods

Useful when approximate solutions are sufficient. This method involves plotting the function and determining the points where it crosses the x-axis. When plotting to the graph the point of intersection of the line with x-axis gives any root of given non-linear equation. E.g. $f(x) = 2x^2 + 5x + 2$. True roots are $-1/2$ and -2 .

Trial and Error

Similar to graphical method, we take a series of guesses for x and then evaluate $f(x)$ at point x . If this returns a value close to 0, then x is one of the approximate root or an error of given non-linear equation. Eg. $f(x) = 2x^2 + x - 6$. True roots are $3/2$ and -2 .

Iterative Methods

This method firstly involves a approximate value of root and then as the algorithm runs for i iterations, more and more accurate estimates of the solution is obtained (which is expected to be converge to the true solution).

Two types of iterative methods based on the number of guesses they use:

1. Bracketing Methods
2. Open end Methods

Bracketing Methods (interpolation methods) start with two initial guesses that 'bracket' the root and then systematically reduce the width of the bracket until the solution is reached. Methods of this category are:

1. Bisection Method
2. False Position Method

Open end methods (extrapolation methods) use a single point value or two values that do not necessarily bracket the roots. Following methods fall under this category:

1. Newton-Raphson method
2. Secant method
3. Fixed-point method

8 Bisection Method OR Half Interval Method

This is the most simplest and the most reliable iterative method for the solution of nonlinear equations. **Statement:** If $f(x)$ is real and continuous in the interval $a < x < b$, and $f(a)$ and $f(b)$ are of opposite signs, that is

$$f(a)f(b) < 0$$

then, there is at least one real root in the interval between a and b . (There may be more than one root in the interval.)

Let $x_1 = a$ and $x_2 = b$. Let us also define another point x_0 to be the midpoint between a and b . That is,

$$x_0 = \frac{x_1 + x_2}{2}$$

Now, there exists the following three conditions:

1. if $f(x_0) = 0$, we have a root at x_0 .

2. if $f(x_0)f(x_1) < 0$, there is root between x_0 and x_1 .
3. if $f(x_0)f(x_2) < 0$, there is a root between x_0 and x_2 .

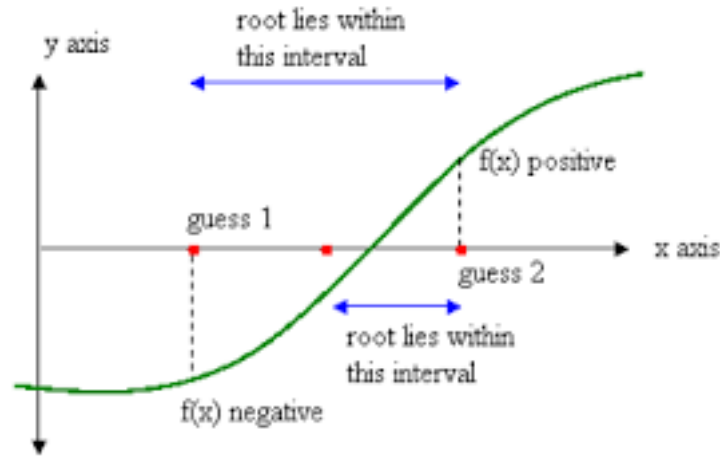


Figure 2: Bisection method illustration

Advantages of Bisection Method:

1. This method is always convergent (brackets the roots).
2. With increasing iterations, the interval gets halved each time. So, error is guaranteed to decrease.

Disadvantages:

1. The convergence of bisection method is slow as it is simply based on halving the interval.
2. If one of the initial guesses is closer to the root, it will take larger number of iterations to reach the root.

Bisection Method Algorithm

Input: Function $f(x)$, initial bounds x_1, x_2 , tolerance ϵ , maximum iterations N .

Output: Approximate root x_{root} or an error message.

Procedure:

1. Decide initial values for x_1, x_2, ϵ , and the maximum number of iterations N .
2. Compute $f_1 \leftarrow f(x_1)$ and $f_2 \leftarrow f(x_2)$.
3. **If** $f_1 \times f_2 > 0$:
 - x_1 and x_2 do not bracket any root. Go to Step 7. **Terminate**.
- Else:**
 - Continue (a root is bracketed).

4. Compute midpoint $x_0 \leftarrow (x_1 + x_2)/2$ and compute $f_0 \leftarrow f(x_0)$. Increment iteration counter $k \leftarrow k + 1$.
 5. **If** $f_1 \times f_0 < 0$:
 - Set $x_2 \leftarrow x_0$.
 - Set $f_2 \leftarrow f_0$.

Else:

 - Set $x_1 \leftarrow x_0$.
 - Set $f_1 \leftarrow f_0$.
 6. **If** $\left| \frac{x_2 - x_1}{x_2} \right| < \epsilon$ **or** $k > N$:
 - $x_{root} \leftarrow (x_1 + x_2)/2$.
 - Write the value of the root x_{root} .
 - Go to Step 7.

Else:

 - Go to Step 4.
 7. **Stop.**
-

Convergence of Bisection Method

In bisection method interval is halved in every interval. After n_{th} iteration, size of iteration is reduced to:

$$\Delta_n = \frac{x_u - x_l}{2^n}$$

Now we can say that maximum error after n_{th} iteration is $E_n = \pm \Delta_n$.

$$|E_n| = \frac{x_u - x_l}{2^n}$$

Similarly, after $(n + 1)^{th}$ iteration maximum error is given by:

$$\begin{aligned} E_{n+1} &= \frac{x_u - x_l}{2^n} \\ E_{n+1} &= \frac{x_u - x_l}{2^n \times 2} \\ E_{n+1} &= \frac{E_n}{2^{n+1}} \end{aligned}$$

Now, the ratio of the error of n^{th} and $(n + 1)^{th}$ iteration is:

$$\frac{E_{n+1}}{E_n} = \frac{\frac{E_n}{2^{n+1}}}{\frac{E_n}{2^n}}$$

This shows that the error decreases by half linearly after each iteration. So, the bisection method is linearly convergent.

Question 1: Find the root of $e^{-x} - x$ upto 3 decimal places. **Ans:0.5673**

Question 2: Find the root of $\cos x = xe^x$ upto 4 decimal places. **Ans:0.51778**

Question 3: Find the root $\sin x = 1/x$ upto 2 decimal places. **Ans:1.113**

Question 4: Find the root $\cos x - 3x + 1 = 0/x$ upto 3 decimal places. **Ans:0.6068**

Question 5: Find the root $2x - \log_1 0x - 7 = 0/x$ upto 4 decimal places. **Ans:3.78927**

Question 6: Find the root $x \log_{10} x = 1.2$ upto 4 decimal places. **Ans:2.7408**

Question 7: Find the root $x^3 - 2x - 5 = 0/x$ with error 0.08% and with error less than 0.00001. **Ans:root=2.0941**

Hint for Q.N.7 The initial guesses for the roots of any polynomial equation can be derived using these formulas:

$$|x^*| = 1 + \frac{1}{|a_n|} \cdot \max(|a_{n-1}|, |a_{n-2}|, \dots, |a_0|)$$

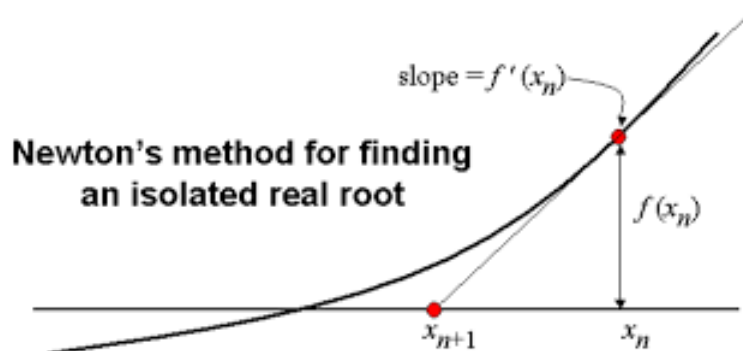
OR

$$|x^*| = \sqrt{\left(\frac{an-1}{a_n}\right)^2 - 2\left(\frac{a_{n-2}}{a_n}\right)}$$

Now, the initial guess interval becomes $(-x^*, x^*)$.

Newton Raphson Method

Newton Raphson is known to be a faster method that calculates the root of the equation. Developed by both Newton and Raphson, this method uses calculus to find the root of any equation whose derivative can be find out.



Derivation

Consider a graph of $f(x)$. Let us assume that x_1 is an approximate root of $f(x) = 0$. Draw a tangent at the curve $f(x)$ at $x = x_1$. Let the point of intersection of the tangent to the x-axis be x_2 . The slope of the tangent is given by:

$$m = f'(x_1)$$

We know that, 2-point equation formula is:

$$y - y_1 = m(x - x_1)$$

Let $y = f(x_2) = 0$, $y_1 = f(x_1)$, $m = f'(x)$ and $x = x_2$. Then,

$$\begin{aligned}
0 - f(x_1) &= f'(x_1)(x_2 - x_1) \\
-f(x_1) &= f'(x_1)(x_2 - x_1) \\
-\frac{f(x_1)}{f'(x_1)} &= (x_2 - x_1) \\
x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)}
\end{aligned}$$

Derivation using the Taylor series

Let h be a small increment in x_n to get the next estimate of the root. Thus,

$$x_{n+1} = x_n + h.$$

We know that Taylor's series can be written as:

$$f(x+h) = f(x) + \frac{f'(x)}{1!}h + \frac{f''(x)}{2!}h^2 + \frac{f'''(x)}{3!}h^3 + \dots$$

Since h is very small, we neglect second and higher-order terms. Thus,

$$f(x+h) = f(x) + f'(x)h.$$

Putting $x = x_n$, we get:

$$f(x_n+h) = f(x_n) + f'(x_n)h.$$

But $x_{n+1} = x_n + h$, so:

$$f(x_{n+1}) = f(x_n) + f'(x_n)(x_{n+1} - x_n).$$

If x_{n+1} is the root of the given equation, then:

$$f(x_{n+1}) = 0.$$

Thus,

$$0 = f(x_n) + f'(x_n)(x_{n+1} - x_n).$$

Solving for x_{n+1} , we obtain:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Algorithm for Newton Raphson's Method:

1. Start
2. Read x , ϵ , n , d .
3. Do for $i = 1$ to n .
4. $f_0 = f(x)$
5. $f_1 = f'(x)$
6. if $(|f_1| < d)$ then display too small slope and goto 11.
7. $x_1 = x_0 - \frac{f_0}{f_1}$
8. if $\left(\left|\frac{x_1 - x_0}{x_1}\right| < \epsilon\right)$ then display root x_1 and goto 11.
9. $x_0 = x_1$ end loop.
10. Display-Method does not converge due to oscillation.
11. Stop.

Disadvantages of NR method

1. Division by zero may occur if $f'(x_i)$ is zero or very close to zero.
2. If the initial guess is too far away from the required root, the process may converge to some other root.
3. A particular value in the iteration sequence may repeat, resulting in an infinite loop. This occurs when the tangent to the curve $f(x)$ at $x = x_{i+1}$ cuts the x-axis again at $x = x_i$.

Convergence of NR method

Let x_n be an estimate of a root of the function $f(x)$. If x_n and x_{n+1} are close to each other, then, using Taylor's series expansion, we can state

$$f(x_{n+1}) = f(x_n) + f'(x_n)(x_{n+1} - x_n) + \frac{f''(R)}{2}(x_{n+1} - x_n)^2$$

where the last term is the remainder term (error term) and R lies somewhere in the interval x_n to x_{n+1} and third and higher order have been dropped.

Let us assume that the exact root of $f(x)$ is x_r . Then $x_{n+1} = x_r$. Therefore $f(x_{n+1}) = 0$ and substituting these values in equation above, we get,

$$0 = f(x_n) + f'(x_n)(x_r - x_n) + \frac{f''(R)}{2}(x_r - x_n)^2 \quad (1)$$

We know that the Newton's iterative formula is given by

$$x_{n+1} = x - \frac{f(x_n)}{f'(x_n)}$$

Rearranging the terms, we get,

$$f(x_n) = f'(x_n)(x_n - x_{n+1})$$

Substituting this for $f(x_n)$ in eq. (1) gives

$$0 = f(x_n) + f'(x_n)(x_r - x_{n+1}) + \frac{f''(R)}{2}(x_r - x_n)^2 \quad (2)$$

We know that the error in the estimate x_{n+1} is given by

$$e_{n+1} = x_r - x_{n+1}$$

Similarly,

$$e_n = x_r - x_n$$

Now, equation (1) can be expressed in terms of these errors as:

$$0 = f'(x_n)e_{n+1} + \frac{f''(R)}{2}e_n^2$$

Rearranging the terms we get,

$$e_{n+1} = -\frac{f''(R)}{2f'(x_n)}e_n^2$$

This shows that the error is roughly proportional to the square of the error in the previous iteration. Therefore, the Newton-Raphson method is said to have quadratic convergence.

Some Deductions

1. $1/N$ is $x_{n+1} = x_n(2 - Nx_n)$

Solⁿ:

Proof. Let $x = \frac{1}{N}$ or $\frac{1}{x} - N = 0$.

Taking $f(x) = \frac{1}{x} - N$ & $f'(x) = -\frac{1}{x^2}$.

Then Newton's formula gives,

$$\begin{aligned} x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\ &= x_n - \frac{\left(\frac{1}{x_n} - N\right)}{-\frac{1}{x_n^2}} \\ &= x_n + \left(\frac{1}{x_n} - N\right) x_n^2 \\ &= x_n + x_n - Nx_n^2 = x_n(2 - Nx_n) \\ \therefore x_{n+1} &= x_n(2 - Nx_n) \end{aligned}$$

□

2. $\sqrt{N}$ is $x_{n+1} = \frac{1}{2} \left(x_n + \frac{N}{x_n} \right)$

Solⁿ:

Proof. Let $x = \sqrt{N}$ or $x^2 - N = 0$.

Taking $f(x) = x^2 - N$ & $f'(x) = 2x$.

Then Newton's formula gives,

$$\begin{aligned} x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\ &= x_n - \frac{x_n^2 - N}{2x_n} \\ &= \frac{1}{2} \left(x_n + \frac{N}{x_n} \right) \\ \therefore x_{n+1} &= \frac{1}{2} \left(x_n + \frac{N}{x_n} \right) \end{aligned}$$

□

3. $1/\sqrt{N}$ is $x_{n+1} = \frac{1}{2} \left(x_n + \frac{1}{Nx_n} \right)$

Solⁿ:

Proof. Let $x = \frac{1}{\sqrt{N}}$ or $x^2 - \frac{1}{N} = 0$.

Taking $f(x) = x^2 - \frac{1}{N}$ & $f'(x) = 2x$.

Then Newton's formula gives,

$$\begin{aligned} x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\ &= x_n - \frac{x_n^2 - \frac{1}{N}}{2x_n} \end{aligned}$$

$$= \frac{1}{2} \left(x_n + \frac{1}{Nx_n} \right)$$

$$\therefore x_{n+1} = \frac{1}{2} \left(x_n + \frac{1}{Nx_n} \right)$$

□

4. $\sqrt[K]{N}$ is $x_{n+1} = \frac{1}{K} \left[(K-1)x_n + \frac{N}{x_n^{K-1}} \right]$

Solⁿ:

Proof. Let $x = \sqrt[K]{N}$ or $x^K - N = 0$.

Taking $f(x) = x^K - N$ & $f'(x) = Kx^{K-1}$.

Then Newton's formula gives,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{x_n^K - N}{Kx_n^{K-1}}$$

$$= \frac{1}{K} \left[(K-1)x_n + \frac{N}{x_n^{K-1}} \right]$$

$$\therefore x_{n+1} = \frac{1}{K} \left[(K-1)x_n + \frac{N}{x_n^{K-1}} \right]$$

□

- **Q.N.1** Find the root of the following equation $x^3 - x^2 + x + 7 = 0$ correct upto 3 decimal places. **Ans:** -1.4883
- **Q.N.2** Find the root of the following equation $x \exp^x - 2 = 0$ correct upto 3 decimal places. **Ans:** 0.8526
- **Q.N.3** Evaluate the value of $\sqrt{2}$ using NR method to 4 decimal places.
- **Q.N.4** Find the root of the following equation $3x - \cos x - 1 = 0$ correct upto 3 decimal places. **Ans:** 0.6071
- **Q.N.5** Find the reciprocal of 3 upto 4 decimal places.

Secant Method

This uses two initial estimates but does not require that they must bracket the root. If the secant method uses the points x_1 and x_2 as starting values, although they do not bracket the root. Slope of the secant line passing through x_1 and x_2 is given by

$$\frac{f(x_1)}{x_1 - x_3} = \frac{f(x_2)}{x_2 - x_3}$$

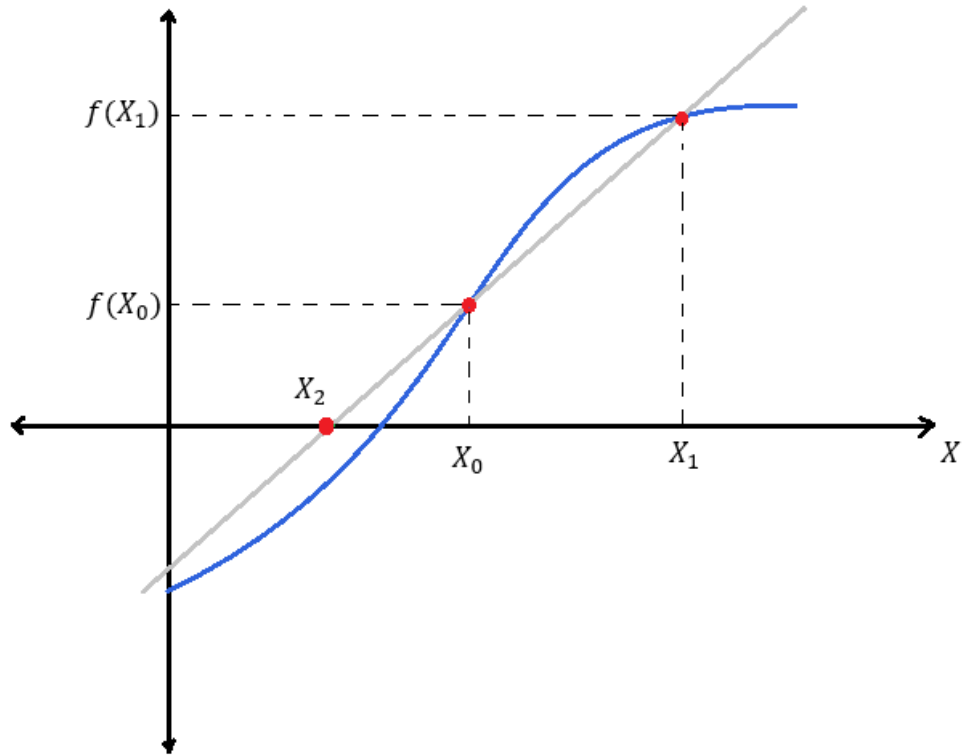
$$f(x_1)(x_2 - x_3) = f(x_2)(x_1 - x_3)$$

$$x_3[f(x_2) - f(x_1)] = f(x_2)x_1 - f(x_1)x_2$$

Then

$$x_3 = \frac{f(x_2)x_1 - f(x_1)x_2}{f(x_2) - f(x_1)}$$

By adding and subtracting $f(x_2)x_2$ to the numerator and rearranging the terms we get



$$x_3 = x_2 - \frac{f(x_2)(x_2 - x_1)}{f(x_2) - f(x_1)}$$

The above equation is known as the *secant formula*. For multiple iterations, its general formula can be written as:

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Algorithm

1. **Start.**
2. Decide two initial points x_1 and x_2 , and ϵ .
3. Compute $f_1 \leftarrow f(x_1)$ and $f_2 \leftarrow f(x_2)$.
4. Compute $x_3 \leftarrow x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$.
5. If $\left| \frac{x_3 - x_2}{x_3} \right| > \epsilon$, then
 - set $x_1 \leftarrow x_2$ and $f_1 \leftarrow f_2$
 - set $x_2 \leftarrow x_3$ and $f_2 \leftarrow f(x_3)$
 - goto step 3

Else:

- set root $\leftarrow x_3$
- print result

6. **Stop.**

Convergence of Secant Method

The secant formula is:

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Let x_r be actual root of $f(x)$ and e_i the error in the estimate of x_i . Then,

$$x_{i+1} = e_{i+1} + x_r$$

$$x_i = e_i + x_r$$

$$x_{i-1} = e_{i-1} + x_r$$

Substituting these in the secant method's formula, and simplifying, we get the error equation as:

$$e_{i+1} = \frac{e_{i-1}f(x_i) - e_i f(x_{i-1})}{f(x_i) - f(x_{i-1})} \quad (a)$$

According to the Mean Value Theorem, there exists at least one point, say $x = R_i$, in the interval x_i and x_r such that,

$$f'(R_i) = \frac{f(x_i) - f(x_r)}{x_i - x_r}$$

We know that

$$f(x_r) = 0$$

$$x_i - x_r = e_i$$

and

$$f'(R_i) = \frac{f(x_i)}{e_i},$$

or

$$f(x_i) = e_i f'(R_i).$$

Similarly,

$$f(x_{i-1}) = e_{i-1} f'(R_{i-1}).$$

Substituting these in the numerator of Eq. (a), we get

$$e_{i+1} = e_i e_{i-1} \left[\frac{f'(R_i) - f'(R_{i-1})}{f(x_i) - f(x_{i-1})} \right].$$

That is, we can say

$$e_{i+1} \propto e_i e_{i-1} \quad (1)$$

We know that the order of convergence of an iteration process is p , if

$$e_i \propto e_{i-1}^p. \quad (2)$$

or

$$e_{i+1} \propto e_i^p. \quad (3)$$

Substituting for e_i and e_{i-1} in Eq. (1), we get

$$e_i^p \propto e_{i-1}^p e_{i-1},$$

or

$$e_i \propto e_{i-1}^{(p+1)/p}. \quad (4)$$

Comparing the relations (2) and (1), we observe that

$$p = (p + 1)p$$

That is,

$$p^2 - p - 1 = 0$$

which has the solutions

$$p = \frac{1 \pm \sqrt{5}}{2}.$$

Since p is always positive, we have

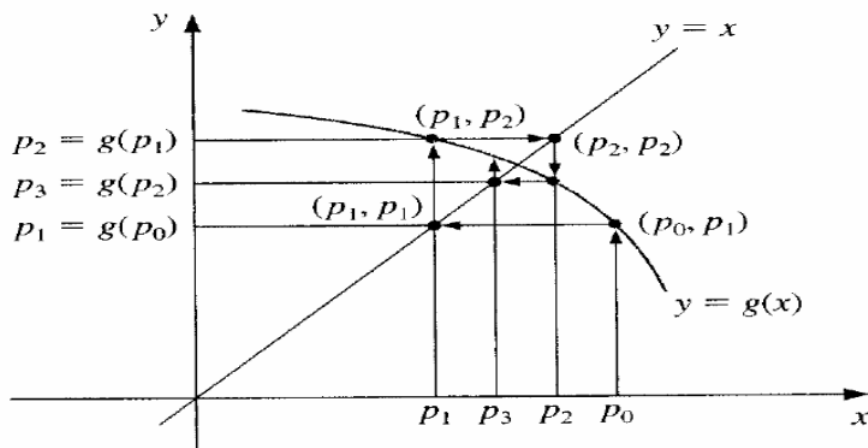
$$p = 1.618.$$

It follows that the order of convergence of the secant method is 1.618 and the convergence is referred to as superlinear convergence.

Q.N.1 Find the root of the equation $xe^x = \cos x$ using the secant method correct upto the four decimal places.

Q.N.2 Find the real root of the equation $3x = \cos x + 1$ using the secant method correct upto the 3 decimal places.

Fixed Point Method



Any function in the form of

$$f(x) = 0$$

can be manipulated such that x is on the left-hand side of the equation as shown below

$$x = g(x)$$

Such an equation is known as the *fixed point equation*. This provides a convenient form for predicting the value of x as a function of x . If x_0 is the initial guess to a root, then the next approximation is given by:

$$x_1 = g(x_0)$$

and so on. This iteration has the general form as:

$$x_{i+1} = g(x_i) \quad i = 0, 1, 2, \dots$$

Algorithm

1. Start.
2. Initialize the value for x_0 and stopping criteria ϵ .
3. Calculate $x_1 = g(x_0)$.
4. If $|x_1 - x_0| < \epsilon$:
 - got step 5.
- Else:
 - $x_0 = x_1$
 - goto step 3.
5. Display x_1 as result i.e. root.
6. Stop.

Convergence of Fixed Point Method

Convergence of this iteration process depends on the nature of $g(x)$. We can theoretically prove this as follows:

The iteration formula is

$$x_{n+1} = g(x_n) \quad (\text{i})$$

Let x_r be the root of given equation then,

$$x_r = g(x_r) \quad (\text{ii})$$

Subtracting above 2 equations we get,

$$x_r - x_{n+1} = g(x_r) - g(x_n) \quad (\text{iii})$$

According to MVT, there is at least one point say $x = R$ in the interval between x_f and x_n such that,

$$g'(R) = \frac{g(x_r) - g(x_n)}{x_r - x_n}$$
$$g(x_r) - g(x_n) = g'(R)(x_r - x_n)$$

Substituting this equation in (iii),

$$x_r - x_{n+1} = g'(R)(x_r - x_n) \quad (\text{iv})$$

If e_i represents the error in the i th iteration, then equation(iv) becomes,

$$e_{n+1} = g'(R)e_n \quad (\text{v})$$

This gives that the error decreases with each iteration only if $g'(R) < 1$. Hence we say that fixed point method converges only under the condition $g'(R) < 1$. i.e. linearly convergent.

Multiple Roots

Synthetic Division and Deflation

A polynomial of degree n can be expressed as

$$p(x) = (x - x_r)q(x)$$

where x_r is a root of the polynomial $p(x)$ and $q(x)$ is the quotient polynomial of degree $n - 1$. Once a root is found, we can use this fact to obtain a lower degree polynomial $q(x)$ by dividing $p(x)$ by $(x - x_r)$ using a process known as synthetic division. The name "synthetic" is used because the quotient polynomial $q(x)$ is obtained without actually performing the division. The activity of reducing the degree of a polynomial is referred to as *deflation*.

If $p(x) = (x - x_r)q(x)$, then

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = (x - x_r)(b_{n-1} x^{n-1} + b_{n-2} x^{n-2} + \dots + b_1 x + b_0)$$

By comparing the coefficient of like powers of x on both sides of equation, we get the following relations between them:

$$\begin{aligned} a_n &= b_{n-1} \\ a_{n-1} &= b_{n-2} - x_r b_{n-1} \\ &\vdots \\ a_1 &= b_0 - x_r b_1 \\ a_0 &= -x_r b_0 \end{aligned}$$

That is,

$$a_i = b_{i-1} - x_r b_i, \quad i = n, n-1, \dots, 0$$

where $b_n = 0$. Then

$$b_{i-1} = a_i + x_r b_i, \quad i = n \dots 1 \quad \text{and} \quad b_n = 0.$$

Remainder Theorem

The remainder theorem states that when a polynomial $f(x)$ is divided by a linear expression $(x - a)$, the remainder is equal to $f(a)$. This allows us to find the remainder of polynomial division by simply evaluating the polynomial at $x = a$, which is much faster than performing long division. For example, the remainder when dividing $f(x) = 3x^2 - 4x + 7$ by $(x - 1)$ is $f(1)$, which is $3(1)^2 - 4(1) + 7 = 6$.

Multiple Roots By Newton's Method

All real roots of polynomials can be found by repeatedly applying Newton's Method and synthetic division. The synthetic division is used to obtain polynomial of degree $n - 1$ from polynomial of degree n . Root of polynomial can be found by using Newton's formula given below and initial guess:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Algorithm

1. Input the degree and coefficient of polynomial (n nad a_j).
2. Input initial guess x_0 and limit E .
While $N > 1$,

(a) Find the root using Newton-Raphson method as:

$$x_r \leftarrow x_0 - \frac{f(x_0)}{f'(x_0)}$$

(b) Divide the polynomial by $x - x_r$ to get the polynomial of degree $n - 1$.

(c) set $x_0 \leftarrow x_1$.

(d) $n \leftarrow n - 1$.

end While.

3. 1st root $\leftarrow -a_0/a_1$.

Horner's Method for Polynomial Evaluation

It is a way to optimize the task of evaluating a polynomial equation. Horner's method (or Horner's scheme) is an algorithm for polynomial evaluation.

The algorithm is based on Horner's rule, in which a polynomial of the following form:

$$P(x) = \sum_{i=0}^n a_i x^i = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_n x^n,$$

where $a_0, a_1, a_2, \dots, a_n$ are real numbers, and we wish to evaluate the polynomial at a point x_0 .

To accomplish this, we rewrite the polynomial in nested multiplication form.

$$P(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n,$$

$$= a_0 + x(a_1 + a_2 x + \cdots + a_n x^{n-1}),$$

$$= a_0 + x\left(a_1 + x(a_2 + \cdots + a_n x^{n-2})\right),$$

$\vdots$

$$P(x) = a_0 + x\left(a_1 + x\left(a_2 + x\left(a_3 + \cdots + x(a_{n-1} + xa_n)\right)\right)\right).$$

Now we define a new sequence of constants as follows:

$$b_n = a_n,$$

$$b_{n-1} = a_{n-1} + b_n x_0,$$

$\vdots$

$$b_0 = a_0 + b_1x_0.$$

Then b_0 is the value of $P(x_0)$.

Algorithm

1. Start.
2. Enter the degree of the polynomial.
3. Enter the value at which the polynomial to be evaluated, x .
4. Initially set $b_n = a_n$.
5. While $n > 0$:

$$b_{n-1} \leftarrow a_{n-1} + b_n * x$$

6. End While.
7. Display the value of b_0 , which is the value of polynomial at x .
8. Stop.